ENG 346
Data Structures and Algorithms for Artificial Intelligence
Matrix Operations and Numpy

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https://github.com/mehmetpekmezci/GTU-ENG-346

ENG-346 Teams code is **0uv7jlm**

Codes

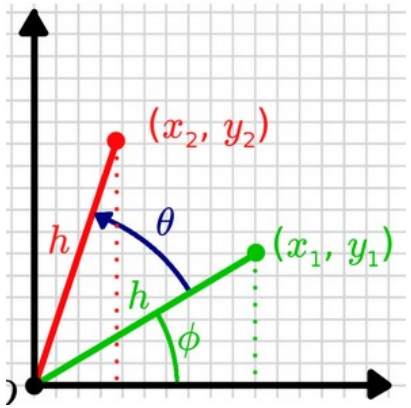
- https://github.com/mehmetpekmezci/GTU-ENG-346/tree/main/matrix-numpy-codes

Matrices

- Matrices can be used to store data
  - Image data (nxm) matrix
  - Sound data (nx1) matrix
  - Any Log data (nxm) matrix,
    - e.g. : log of humidity, temperature, … values in a room
  - Geometric data
- Matrices can be used for Linear Transformations (Linear Function)
  - Rotation
  - Translation
  - Stretching

Matrices – Geometric Meaning

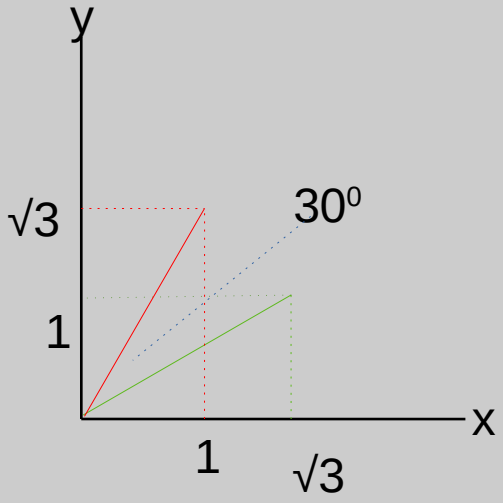
- **Rotation**



$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

NOTE: 3D rotation matrix is a 3x3 matrix, not a 2x2x2 tensor

Example :



$$\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} = \begin{pmatrix} \cos(\pi/6) & -\sin(\pi/6) \\ \sin(\pi/6) & \cos(\pi/6) \end{pmatrix} \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}$$

```
import numpy as np
theta = np.radians(30)
c, s = np.cos(theta), np.sin(theta)
R = np.array(((c, -s), (s, c)))
green_vector=np.array([np.sqrt(3),1])
print(green_vector)
red_vector=np.dot(R,green_vector)
print(red_vector)
```

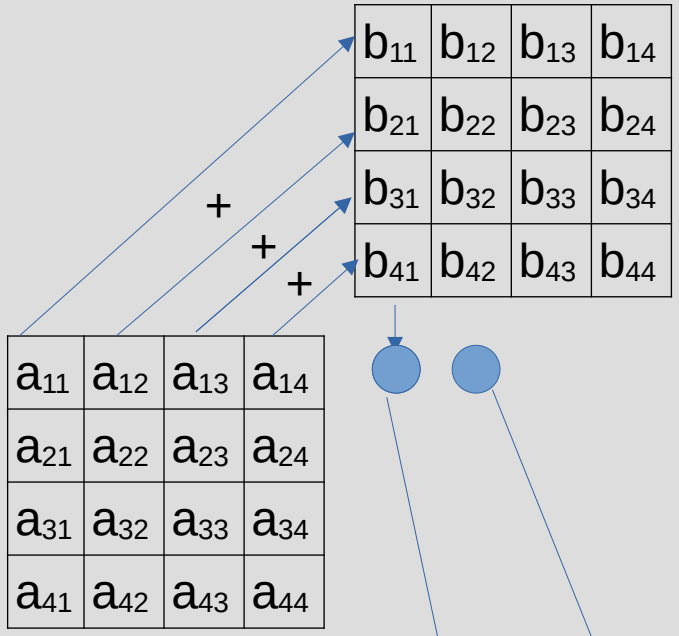
https://articulatedrobotics.xyz/tutorials/coordinate-transforms/rotation-matrices-2d/

Multiplication of Matrices

Example : 4x4 matrix A=

- We can multiply A and B If A(N,**M**) . B(**M**,K)
- A(**N**,M). B(M,**K**) = C(**N,K**)

A.B =



a11.b12+ a12.b22+a13.b32+a14.b42

a11.b11+ a12.b21+a13.b31+a14.b41

Matrix Example

- Example :

$$x + 2y = 4$$
$$3x - 4y = 5$$

$$\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$A \quad . X \quad = \quad B$$

$$\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ + \; y \end{bmatrix} \begin{bmatrix} x+2y \\ 3x-4y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \Rightarrow$$

$$x + 2y = 4$$
$$3x - 4y = 5$$

Determinants and Inverse of a Matrix

Example : 4x4 matrix A=

a	b	c	d
e	f	g	h
i	j	k	l
m	n	o	p

Minor of element b is det(M_{12}),

Cofactor(A) is a matrix composed of C_{ij} .

$C_{ij} = (-1)^{i+j} \det(M_{ij})$

M_{12} =

e	g	h
i	k	l
m	o	p

det(A)=|A| = **a . det (**

	f	g	h
	j	k	l
	n	o	p

) - b . det (

e		g	h
i		k	l
m		o	p

)

+ c . det (

e	f		h
i	j		l
m	n		p

) - d . det(

e	f	g	
i	j	k	
m	n	o	

)

C_{11}	C_{12}	C_{13}	C_{14}
C_{21}	C_{22}	C_{23}	C_{24}
C_{31}	C_{32}	C_{33}	C_{34}
C_{41}	C_{42}	C_{43}	C_{44}

Inverse of A

$A^{-1} = 1/\det(A).\ (\text{Cofactor}(A))^T$

NO INVERSE

if det(A)=0 or

matrix is not square

Transpose and Inverse Properties

$A \,.\, A^{-1} = A^{-1} \,.\, A = I$

$(A^{-1})^{-1} = A$

$(A^T)^T = A$

$(A^T)^{-1} = (A^{-1})^T$

$(AB)^T = B^T \,.\, A^T$

$(A+B)^T = A^T + B^T$

$(AB)^{-1} = B^{-1} \,.\, A^{-1}$

$(k.A)^{-1} = 1/k \,.\, A^{-1}$

$(k.A)^T = k \,.\, A^T$

$\det(A^T) = \det(A)$

$\det(A^{-1}) = 1/\det(A)$

$\det(A.B)=\det(A).\det(B)$

Tensors (Multilinear Map)

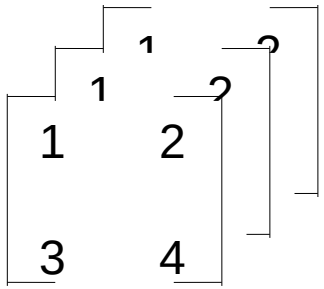
- Vectors = 1st order Tensor $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (1D arrays)

- Matrices = 2nd order Tensor $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (2D arrays)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

3x3 matrix Represented as 2D array data structure.

- 3rd order Tensor , is a stack of matrices (3D arrays)



→ **2 x 2 x 3** Tensor (I ,j, k indices)

- Tensor may have n order.

Matrix Operations

- Matrix/Tensor Construction
- Dimensions (Shape) and reshaping, array slicing, masking.
- Matrix Summation
- Transpose of the Matrix
- Matrix Multiplication
- Inverse of the Matrix
- Stacking (Vertical, Horizontal, Depth)
- Splitting (Vertical, Horizontal)
- Distances (Mean Squared Error, Mean Absolute Error, Cosine Similarity)
- **NOT COVERED : Eigenvalues, Eigenvectors, SVD , Tensor Dot**

Matrix/Tensor Construction

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \longleftrightarrow$$

```
import numpy as np
a = np.array( [   [1, 2] , [3, 4] ] )
print(a)
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 01.arrays_matrices.py
[[1 2]
 [3 4]]
```

$$\begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix}$$

```
np.full((2, 2), 10,dtype=int)
# 2 rows, 2 columns
```

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

```
number_of_rows=2
np.identity(number_of_rows,dtype=int)
```

$$\begin{bmatrix} 0.0 & 0.0 \\ 0.0 & 0.0 \end{bmatrix}$$

```
np.zeros(2)
# default dtype=float
```

$$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$$

```
number_of_rows=2
np.identity(number_of_rows,dtype=float)
```

Matrix/Tensor Construction

```
import numpy as np
np.random.seed(0)  # seed for reproducibility
random_tensor = np.random.randint(100, size=(3, 4, 5))
print(random_tensor)
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 02.random.py
[[[44 47 64 67 67]
  [ 9 83 21 36 87]
  [70 88 88 12 58]
  [65 39 87 46 88]]

 [[81 37 25 77 72]
  [ 9 20 80 69 79]
  [47 64 82 99 88]
  [49 29 19 19 14]]

 [[39 32 65  9 57]
  [32 31 74 23 35]
  [75 55 28 34  0]
  [ 0 36 53  5 38]]]
```

Matrix Dimensions

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

print("\nRESHAPED TO (2,3) ARRAY")
b=a.reshape((2,3))
print(f"MATRIX DIM : {b.shape}")
print(f"MATRIX : {b}")


print("\nRESHAPED TO (6,1) ARRAY")
c=a.reshape((6,))
print(f"MATRIX DIM : {c.shape}")
print(f"MATRIX : {c}")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 03.matrix.dim.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

RESHAPED TO (2,3) ARRAY
MATRIX DIM : (2, 3)
MATRIX : [[1 2 3]
 [4 5 6]]

RESHAPED TO (6,1) ARRAY
MATRIX DIM : (6,)
MATRIX : [1 2 3 4 5 6]
```

Array Slicing

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2,11] , [3, 4,12] , [5,6,13] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")


b=np.copy(a[1,:2])
print(f"\nMATRIX DIM : {b.shape}")
print(f"MATRIX : {b}")

a[1:,1]=8
print(f"\nMATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

a[2:,1:]=b
print(f"\nMATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")
```

```
mpekmezci@cobalt:~$ python3 04.array.slicing.py
INITIAL ARRAY
MATRIX DIM : (3, 3)
MATRIX : [[ 1  2 11]
 [ 3  4 12]
 [ 5  6 13]]

MATRIX DIM : (2,)
MATRIX : [3 4]

MATRIX DIM : (3, 3)
MATRIX : [[ 1  2 11]
 [ 3  8 12]
 [ 5  8 13]]

MATRIX DIM : (3, 3)
MATRIX : [[ 1  2 11]
 [ 3  8 12]
 [ 5  3  4]]
```

Array Masking

```
import numpy as np

print("INITIAL ARRAY")
a = np.arange(15)
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

wanted_indices=[3,8,12]

b=np.copy(a[wanted_indices])
print(f"\nMATRIX DIM : {b.shape}")
print(f"MATRIX : {b}")
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 05.array.masking.py
INITIAL ARRAY
MATRIX DIM : (15,)
MATRIX : [ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14]

MATRIX DIM : (3,)
MATRIX : [ 3  8 12]
```

Matrix Summation

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=a+a
print(f"\n MATRIX:{b} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 06.summation.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

 MATRIX:[[ 2  4]
 [ 6  8]
 [10 12]]
```

Matrix Transpose

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=a.T
print(f"\n MATRIX:{b} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ vi 07.transpose.py
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 07.transpose.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

 MATRIX:[[1 3 5]
 [2 4 6]]
```

Matrix Multiplication

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=np.matmul(a.T , a)
print(f"\n MATRIX:{b} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 08.multiplication.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

 MATRIX:[[35 44]
 [44 56]]
```

Matrix Inverse

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1,2,3] , [5,7,11] , [13,17,23] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=np.linalg.inv(a)
print(f"\n INVERSE MATRIX:{b} \n")


c=np.matmul(b,a)
print(f"\n A INVERSE * A :{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 5 & 7 & 11 \\ 13 & 17 & 23 \end{bmatrix}$$

Very close to **0**

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 09.inverse.
INITIAL ARRAY
MATRIX DIM : (3, 3)
MATRIX : [[ 1  2  3]
 [ 5  7 11]
 [13 17 23]]

 INVERSE MATRIX:[[-2.16666667  0.41666667  0.08333333]
 [ 2.33333333 -1.33333333  0.33333333]
 [-0.5         0.75       -0.25      ]]


 A INVERSE * A :[[ 1.00000000e+00  1.08246745e-15  1.58206781e-15]
 [-4.99600361e-16  1.00000000e+00 -6.10622664e-16]
 [ 2.77555756e-16  9.43689571e-16  1.00000000e+00]]
```

Horizontal Stacking

```
import numpy as np

a = np.array( [   [1, 2,3] , [4, 5, 6 ] , [7,8,9] ] )
print(f"MATRIX A : {a}")

b = np.array( [   [11, 12,13] , [14, 15, 16 ] , [17,18,19] ] )
print(f"MATRIX B : {b}")

c=np.hstack((a,b))
print(f"\n HSTACK:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \qquad \begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 10.hstack.py
MATRIX A : [[1 2 3]
 [4 5 6]
 [7 8 9]]
MATRIX B : [[11 12 13]
 [14 15 16]
 [17 18 19]]

 HSTACK:[[ 1  2  3 11 12 13]
 [ 4  5  6 14 15 16]
 [ 7  8  9 17 18 19]]
```

Vertical Stacking

```
import numpy as np

a = np.array( [   [1, 2,3] , [4, 5, 6 ] , [7,8,9] ] )
print(f"MATRIX A : {a}")

b = np.array( [   [11, 12,13] , [14, 15, 16 ] , [17,18,19] ] )
print(f"MATRIX B : {b}")

c=np.vstack((a,b))
print(f"\n VSTACK:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \qquad \begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 11.vstack.py
MATRIX A : [[1 2 3]
 [4 5 6]
 [7 8 9]]
MATRIX B : [[11 12 13]
 [14 15 16]
 [17 18 19]]

 VSTACK:[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [11 12 13]
 [14 15 16]
 [17 18 19]]
```

Depth Stacking

```
import numpy as np

a = np.array( [   [1, 2,3] , [4, 5, 6 ] , [7,8,9] ] )
print(f"MATRIX A : {a}")

b = np.array( [   [11, 12,13] , [14, 15, 16 ] , [17,18,19] ] )
print(f"MATRIX B : {b}")

c=np.dstack((a,b))
print(f"\n DSTACK:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \qquad \begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 12.dstack.py
MATRIX A : [[1 2 3]
 [4 5 6]
 [7 8 9]]
MATRIX B : [[11 12 13]
 [14 15 16]
 [17 18 19]]

 DSTACK:[[[ 1 11]
  [ 2 12]
  [ 3 13]]

 [[ 4 14]
  [ 5 15]
  [ 6 16]]

 [[ 7 17]
  [ 8 18]
  [ 9 19]]]
```

Horizontal Splitting

```
import numpy as np

a = np.array( [   [1, 2,3,4] , [5, 6, 7,8 ] ] )
print(f"MATRIX A : {a}")

b,c=np.hsplit(a,2)
print(f"\n First Split:{b} \n")
print(f"\n Second Split:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 13.hsplit.py
MATRIX A : [[1 2 3 4]
 [5 6 7 8]]

 First Split:[[1 2]
 [5 6]]


 Second Split:[[3 4]
 [7 8]]
```

Vertical Splitting

```
import numpy as np

a = np.array( [   [1, 2] ,[3,4] , [5, 6], [7,8 ] ] )
print(f"MATRIX A : {a}")

b,c=np.vsplit(a,2)
print(f"\n First Split:{b} \n")
print(f"\n Second Split:{c} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 14.vsplit.py
MATRIX A : [[1 2]
 [3 4]
 [5 6]
 [7 8]]

 First Split:[[1 2]
 [3 4]]


 Second Split:[[5 6]
 [7 8]]
```

Distance – Root Mean Squared Error – Euclidean Distance

```
import numpy as np

a = np.array( [   [1, 2] ,[3,4]  ] )
print(f"MATRIX A : {a}")

b = np.array( [   [5, 6] ,[7,8]  ] )
print(f"MATRIX B : {b}")

rmse = np.sqrt((np.square(a - b)).mean())
print(rmse)
```

$$A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B=\begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$$

$$RMSE(A,B)=\left(\frac{1}{N*M}\sum_{i=0}^{N}\sum_{j=0}^{M}(a_{ij}-b_{ij})^2\right)^{\frac{1}{2}}$$

N=2, M=2 in this case

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 15.rmse.py
MATRIX A : [[1 2]
 [3 4]]
MATRIX B : [[5 6]
 [7 8]]
4.0
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$
```

Distance – Mean Absolute Error

```
import numpy as np

a = np.array( [   [1, 2] ,[3,4]  ] )
print(f"MATRIX A : {a}")

b = np.array( [   [5, 6] ,[7,8]  ] )
print(f"MATRIX B : {b}")

mae=np.absolute(a-b).mean()
print(mae)
```

$$A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B=\begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$$

$$MAE(A,B)=\frac{1}{N*M}\sum_{i=0}^{N}\sum_{j=0}^{M}|a_{ij}-b_{ij}|$$

N=2, M=2
in this case

```
print(mse)
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 16.mae.py
MATRIX A : [[1 2]
 [3 4]]
MATRIX B : [[5 6]
 [7 8]]
4.0
```

Distance – Cosine Distance

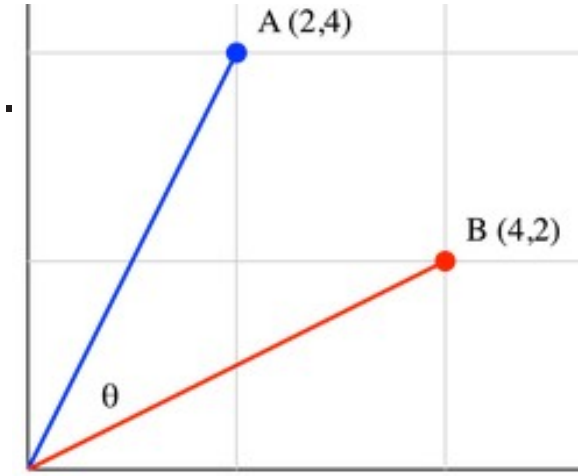
```
import numpy as np
from numpy.linalg import norm

A = np.array([2, 4])
B = np.array([4, 2])

# compute cosine similarity
cosine = np.dot(A, B) / (norm(A) * norm(B))
print("Cosine Distance:", 1-cosine)
```

$$COSINEDISTANCE(A,B)=1-COSINESIMILARITY(A,B)$$

$$COSINESIMILARITY(A,B)=\frac{A.B}{\|A\|.\|B\|}=\cos(\hat{AB})=\cos(\theta)$$

$$\|A\|=Euclidean(L2)\,Norm\,of\;A=\left(\sum_{i=0}^{N}\sum_{j=0}^{M}(a_{ij})^2\right)^{\frac{1}{2}}$$

$$A=\begin{bmatrix}2\\4\end{bmatrix}\quad B=\begin{bmatrix}4\\2\end{bmatrix}$$




```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 17.cosine.py
Cosine Distance: 0.20000000000000018
```

Distance – Cosine Distance - Matrix

```
import numpy as np
from numpy.linalg import norm

A = np.array([[1, 2, 2], [3, 2, 2], [-2, 1, -3]])
B = np.array([[4, 2, 4], [2, -2, 5], [3, 4, -4]])

cosine = np.sum(A * B, axis=1) / (norm(A, axis=1) * norm(B,
axis=1))
print("Cosine Similarity:", cosine)
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 17.cosine.matrix.py
Cosine Similarity: [0.88888889 0.5066404  0.41739194]
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$
```

ENG 346
Data Structures and Algorithms for Artificial Intelligence
Matrix Operations and Numpy

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https://github.com/mehmetpekmezci/GTU-ENG-346

ENG-346 Teams code is **0uv7jlm**

Codes

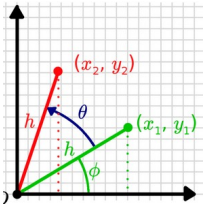
- https://github.com/mehmetpekmezci/GTU-ENG-346/tree/main/matrix-numpy-codes

Matrices

- Matrices can be used to store data
  - Image data (nxm) matrix
  - Sound data (nx1) matrix
  - Any Log data (nxm) matrix,
    - e.g. : log of humidity, temperature, … values in a room
  - Geometric data
- Matrices can be used for Linear Transformations (Linear Function)
  - Rotation
  - Translation
  - Stretching

Matrices – Geometric Meaning

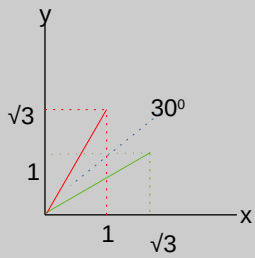
- **Rotation**



$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

NOTE: 3D rotation matrix is a 3x3 matrix, not a 2x2x2 tensor

Example :



$$\begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} = \begin{bmatrix} \cos(\pi/6) & -\sin(\pi/6) \\ \sin(\pi/6) & \cos(\pi/6) \end{bmatrix} \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}$$

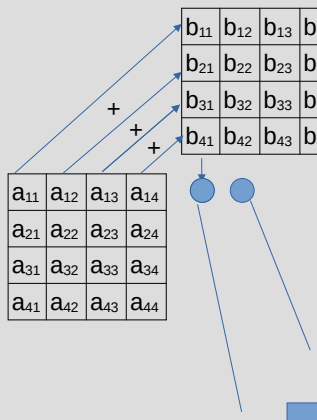
```
import numpy as np
theta = np.radians(30)
c, s = np.cos(theta), np.sin(theta)
R = np.array(((c, -s), (s, c)))
green_vector=np.array([np.sqrt(3),1])
print(green_vector)
red_vector=np.dot(R,green_vector)
print(red_vector)
```

https://articulatedrobotics.xyz/tutorials/coordinate-transforms/rotation-matrices-2d/

Multiplication of Matrices

Example : 4x4 matrix A=

A.B =



- We can multiply A and B If A(N,**M**) . B(**M**,K)
- A(**N**,M). B(M,**K**) = C(**N,K**)

a11.b12+ a12.b22+a13.b32+a14.b42

a11.b11+ a12.b21+a13.b31+a14.b41

Matrix Example

- Example :

$$x + 2y = 4$$
$$3x - 4y = 5$$

$$\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$A \quad . X \quad = \quad B$$

$$\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ + \; y \end{bmatrix} \begin{bmatrix} x+2y \\ 3x-4y \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \Rightarrow \begin{matrix} x + 2y = 4 \\ 3x - 4y = 5 \end{matrix}$$

Determinants and Inverse of a Matrix

Example : 4x4 matrix A=

a	b	c	d
e	f	g	h
i	j	k	l
m	n	o	p

Minor of element b is det(M_{12}), **M_{12} =**

e	g	h
i	k	l
m	o	p

Cofactor(A) is a matrix composed of C_{ij} .

$C_{ij} = (-1)^{i+j} \det(M_{ij})$

det(A)=|A| = **a . det (**

	f	g	h
	j	k	l
	n	o	p

) - b . det (

e		g	h
i		k	l
m		o	p

)

+ c . det (

e	f		h
i	j		l
m	n		p

) - d . det(

e	f	g	
i	j	k	
m	n	o	

)

C_{11}	C_{12}	C_{13}	C_{14}
C_{21}	C_{22}	C_{23}	C_{24}
C_{31}	C_{32}	C_{33}	C_{34}
C_{41}	C_{42}	C_{43}	C_{44}

Inverse of A

$A^{-1} = 1/\det(A).\ (\text{Cofactor}(A))^T$

NO INVERSE
if det(A)=0 or
matrix is not square

Transpose and Inverse Properties

$A \cdot A^{-1} = A^{-1} \cdot A = I$

$(A^{-1})^{-1} = A$

$(A^T)^T = A$

$(A^T)^{-1} = (A^{-1})^T$

$(AB)^T = B^T \cdot A^T$

$(A+B)^T = A^T + B^T$

$(AB)^{-1} = B^{-1} \cdot A^{-1}$

$(k.A)^{-1} = 1/k \cdot A^{-1}$

$(k.A)^T = k \cdot A^T$

$\det(A^T) = \det(A)$

$\det(A^{-1}) = 1/\det(A)$

$\det(A.B)=\det(A).\det(B)$

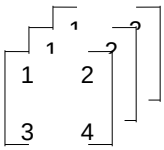
Tensors (Multilinear Map)

- Vectors = 1st order Tensor $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (1D arrays)
- Matrices = 2nd order Tensor $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (2D arrays)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

3x3 matrix Represented as 2D array data structure.

- 3rd order Tensor , is a stack of matrices (3D arrays)



→ **2 x 2 x 3** Tensor (I ,j, k indices)

- Tensor may have n order.

Matrix Operations

- Matrix/Tensor Construction
- Dimensions (Shape) and reshaping, array slicing, masking.
- Matrix Summation
- Transpose of the Matrix
- Matrix Multiplication
- Inverse of the Matrix
- Stacking (Vertical, Horizontal, Depth)
- Splitting (Vertical, Horizontal)
- Distances (Mean Squared Error, Mean Absolute Error, Cosine Similarity)
- **NOT COVERED : Eigenvalues, Eigenvectors, SVD , Tensor Dot**

Matrix/Tensor Construction

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \longleftrightarrow$$

```
import numpy as np
a = np.array( [   [1, 2] , [3, 4] ] )
print(a)
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 01.arrays_matrices.py
[[1 2]
 [3 4]]
```

$$\begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix}$$

```
np.full((2, 2), 10,dtype=int)
# 2 rows, 2 columns
```

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

```
number_of_rows=2
np.identity(number_of_rows,dtype=int)
```

$$\begin{bmatrix} 0.0 & 0.0 \\ 0.0 & 0.0 \end{bmatrix}$$

```
np.zeros(2)
# default dtype=float
```

$$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$$

```
number_of_rows=2
np.identity(number_of_rows,dtype=float)
```

Matrix/Tensor Construction

```
import numpy as np
np.random.seed(0)  # seed for reproducibility
random_tensor = np.random.randint(100, size=(3, 4, 5))
print(random_tensor)
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 02.random.py
[[[44 47 64 67 67]
  [ 9 83 21 36 87]
  [70 88 88 12 58]
  [65 39 87 46 88]]

 [[81 37 25 77 72]
  [ 9 20 80 69 79]
  [47 64 82 99 88]
  [49 29 19 19 14]]

 [[39 32 65  9 57]
  [32 31 74 23 35]
  [75 55 28 34  0]
  [ 0 36 53  5 38]]]
```

Matrix Dimensions

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

print("\nRESHAPED TO (2,3) ARRAY")
b=a.reshape((2,3))
print(f"MATRIX DIM : {b.shape}")
print(f"MATRIX : {b}")


print("\nRESHAPED TO (6,1) ARRAY")
c=a.reshape((6,))
print(f"MATRIX DIM : {c.shape}")
print(f"MATRIX : {c}")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 03.matrix.dim.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

RESHAPED TO (2,3) ARRAY
MATRIX DIM : (2, 3)
MATRIX : [[1 2 3]
 [4 5 6]]

RESHAPED TO (6,1) ARRAY
MATRIX DIM : (6,)
MATRIX : [1 2 3 4 5 6]
```

Array Slicing

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2,11] , [3, 4,12] , [5,6,13] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")


b=np.copy(a[1,:2])
print(f"\nMATRIX DIM : {b.shape}")
print(f"MATRIX : {b}")

a[1:,1]=8
print(f"\nMATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

a[2:,1:]=b
print(f"\nMATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")
```

```
mpekmezci@cobalt:~$ python3 04.array.slicing.py
INITIAL ARRAY
MATRIX DIM : (3, 3)
MATRIX : [[ 1  2 11]
 [ 3  4 12]
 [ 5  6 13]]

MATRIX DIM : (2,)
MATRIX : [3 4]

MATRIX DIM : (3, 3)
MATRIX : [[ 1  2 11]
 [ 3  8 12]
 [ 5  8 13]]

MATRIX DIM : (3, 3)
MATRIX : [[ 1  2 11]
 [ 3  8 12]
 [ 5  3  4]]
```

Array Masking

```
import numpy as np

print("INITIAL ARRAY")
a = np.arange(15)
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

wanted_indices=[3,8,12]

b=np.copy(a[wanted_indices])
print(f"\nMATRIX DIM : {b.shape}")
print(f"MATRIX : {b}")
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 05.array.masking.py
INITIAL ARRAY
MATRIX DIM : (15,)
MATRIX : [ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14]

MATRIX DIM : (3,)
MATRIX : [ 3  8 12]
```

Matrix Summation

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=a+a
print(f"\n MATRIX:{b} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 06.summation.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

 MATRIX:[[ 2  4]
 [ 6  8]
 [10 12]]
```

Matrix Transpose

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=a.T
print(f"\n MATRIX:{b} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ vi 07.transpose.py
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 07.transpose.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

 MATRIX:[[1 3 5]
 [2 4 6]]
```

Matrix Multiplication

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1, 2] , [3, 4] , [5,6] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=np.matmul(a.T , a)
print(f"\n MATRIX:{b} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 08.multiplication.py
INITIAL ARRAY
MATRIX DIM : (3, 2)
MATRIX : [[1 2]
 [3 4]
 [5 6]]

 MATRIX:[[35 44]
 [44 56]]
```

Matrix Inverse

```
import numpy as np

print("INITIAL ARRAY")
a = np.array( [   [1,2,3] , [5,7,11] , [13,17,23] ] )
print(f"MATRIX DIM : {a.shape}")
print(f"MATRIX : {a}")

b=np.linalg.inv(a)
print(f"\n INVERSE MATRIX:{b} \n")


c=np.matmul(b,a)
print(f"\n A INVERSE * A :{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 5 & 7 & 11 \\ 13 & 17 & 23 \end{bmatrix}$$

Very close to **0**

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 09.inverse.
INITIAL ARRAY
MATRIX DIM : (3, 3)
MATRIX : [[ 1  2  3]
 [ 5  7 11]
 [13 17 23]]

 INVERSE MATRIX:[[-2.16666667  0.41666667  0.08333333]
 [ 2.33333333 -1.33333333  0.33333333]
 [-0.5         0.75       -0.25      ]]


 A INVERSE * A :[[ 1.00000000e+00  1.08246745e-15  1.58206781e-15]
 [-4.99600361e-16  1.00000000e+00 -6.10622664e-16]
 [ 2.77555756e-16  9.43689571e-16  1.00000000e+00]]
```

Horizontal Stacking

```
import numpy as np

a = np.array( [   [1, 2,3] , [4, 5, 6 ] , [7,8,9] ] )
print(f"MATRIX A : {a}")

b = np.array( [   [11, 12,13] , [14, 15, 16 ] , [17,18,19] ] )
print(f"MATRIX B : {b}")

c=np.hstack((a,b))
print(f"\n HSTACK:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 10.hstack.py
MATRIX A : [[1 2 3]
 [4 5 6]
 [7 8 9]]
MATRIX B : [[11 12 13]
 [14 15 16]
 [17 18 19]]

 HSTACK:[[ 1  2  3 11 12 13]
 [ 4  5  6 14 15 16]
 [ 7  8  9 17 18 19]]
```

Vertical Stacking

```
import numpy as np

a = np.array( [   [1, 2,3] , [4, 5, 6 ] , [7,8,9] ] )
print(f"MATRIX A : {a}")

b = np.array( [   [11, 12,13] , [14, 15, 16 ] , [17,18,19] ] )
print(f"MATRIX B : {b}")

c=np.vstack((a,b))
print(f"\n VSTACK:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 11.vstack.py
MATRIX A : [[1 2 3]
 [4 5 6]
 [7 8 9]]
MATRIX B : [[11 12 13]
 [14 15 16]
 [17 18 19]]

 VSTACK:[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [11 12 13]
 [14 15 16]
 [17 18 19]]
```

Depth Stacking

```
import numpy as np

a = np.array( [   [1, 2,3] , [4, 5, 6 ] , [7,8,9] ] )
print(f"MATRIX A : {a}")

b = np.array( [   [11, 12,13] , [14, 15, 16 ] , [17,18,19] ] )
print(f"MATRIX B : {b}")

c=np.dstack((a,b))
print(f"\n DSTACK:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \\ 17 & 18 & 19 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 12.dstack.py
MATRIX A : [[1 2 3]
 [4 5 6]
 [7 8 9]]
MATRIX B : [[11 12 13]
 [14 15 16]
 [17 18 19]]

 DSTACK:[[[ 1 11]
  [ 2 12]
  [ 3 13]]

 [[ 4 14]
  [ 5 15]
  [ 6 16]]

 [[ 7 17]
  [ 8 18]
  [ 9 19]]]
```

Horizontal Splitting

```
import numpy as np

a = np.array( [   [1, 2,3,4] , [5, 6, 7,8 ] ] )
print(f"MATRIX A : {a}")

b,c=np.hsplit(a,2)
print(f"\n First Split:{b} \n")
print(f"\n Second Split:{c} \n")
```

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 13.hsplit.py
MATRIX A : [[1 2 3 4]
 [5 6 7 8]]

 First Split:[[1 2]
 [5 6]]


 Second Split:[[3 4]
 [7 8]]
```

Vertical Splitting

```
import numpy as np

a = np.array( [   [1, 2] ,[3,4] , [5, 6], [7,8 ] ] )
print(f"MATRIX A : {a}")

b,c=np.vsplit(a,2)
print(f"\n First Split:{b} \n")
print(f"\n Second Split:{c} \n")
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}$$

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 14.vsplit.py
MATRIX A : [[1 2]
 [3 4]
 [5 6]
 [7 8]]

 First Split:[[1 2]
 [3 4]]


 Second Split:[[5 6]
 [7 8]]
```

Distance – Root Mean Squared Error – Euclidean Distance

```
import numpy as np

a = np.array( [   [1, 2] ,[3,4]  ] )
print(f"MATRIX A : {a}")

b = np.array( [   [5, 6] ,[7,8]  ] )
print(f"MATRIX B : {b}")

rmse = np.sqrt((np.square(a - b)).mean())
print(rmse)
```

$$A=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B=\begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$$

$$RMSE(A,B)=\left(\frac{1}{N*M}\sum_{i=0}^{N}\sum_{j=0}^{M}(a_{ij}-b_{ij})^2\right)^{\frac{1}{2}}$$

N=2, M=2
in this case

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 15.rmse.py
MATRIX A : [[1 2]
 [3 4]]
MATRIX B : [[5 6]
 [7 8]]
4.0
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$
```

Distance – Mean Absolute Error

```
import numpy as np

a = np.array( [   [1, 2] ,[3,4]  ] )
print(f"MATRIX A : {a}")

b = np.array( [   [5, 6] ,[7,8]  ] )
print(f"MATRIX B : {b}")

mae=np.absolute(a-b).mean()
print(mae)
```

A= $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ B= $\begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$

$$MAE(A,B)=\frac{1}{N*M}\sum_{i=0}^{N}\sum_{j=0}^{M}|a_{ij}-b_{ij}|$$

N=2, M=2
in this case

```
print(mse)
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 16.mae.py
MATRIX A : [[1 2]
 [3 4]]
MATRIX B : [[5 6]
 [7 8]]
4.0
```

Distance – Cosine Distance

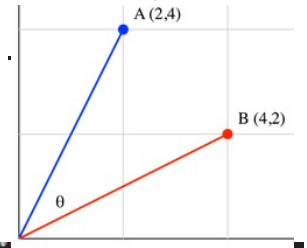
```
import numpy as np
from numpy.linalg import norm

A = np.array([2, 4])
B = np.array([4, 2])

# compute cosine similarity
cosine = np.dot(A, B) / (norm(A) * norm(B))
print("Cosine Distance:", 1-cosine)
```

$$COSINEDISTANCE(A,B)=1-COSINESIMILARITY(A,B)$$

$$COSINESIMILARITY(A,B)=\frac{A.B}{\|A\|.\|B\|}=\cos(\hat{AB})=\cos(\theta)$$

$$\|A\|=Euclidean(L2)\,Norm\,of\;A=\left(\sum_{i=0}^{N}\sum_{j=0}^{M}(a_{ij})^2\right)^{\frac{1}{2}}$$

$$A=\begin{bmatrix}2\\4\end{bmatrix}\quad B=\begin{bmatrix}4\\2\end{bmatrix}$$




```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 17.cosine.py
Cosine Distance: 0.20000000000000018
```

Distance – Cosine Distance - Matrix

```
import numpy as np
from numpy.linalg import norm

A = np.array([[1, 2, 2], [3, 2, 2], [-2, 1, -3]])
B = np.array([[4, 2, 4], [2, -2, 5], [3, 4, -4]])

cosine = np.sum(A * B, axis=1) / (norm(A, axis=1) * norm(B,
axis=1))
print("Cosine Similarity:", cosine)
```

```
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$ python3 17.cosine.matrix.py
Cosine Similarity: [0.88888889 0.5066404  0.41739194]
mpekmezci@cobalt:~/workspace/GTU-ENG-346-PRIVATE/matrix-numpy-codes$
```